

Semester Project Proposal

**Information & Security Lab
CS-251L**



:Project Title:

SpaceCom Defender

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1. Abstract

Space mission control systems depend on highly secure communication channels to send commands and receive telemetry from spacecraft. Any cyberattack targeting these channels—such as command spoofing, unauthorized access, or data tampering—can lead to mission failure. Therefore, ensuring confidentiality, integrity, and availability of mission data is critical.

The objective of **SpaceCom Defender** is to develop a simplified, Python-based cybersecurity simulation that demonstrates how a mission control system can authenticate users, protect sensitive command data using encryption, detect unauthorized access, and maintain secure activity logs. To achieve this, we will implement secure login (hashing), role-based access control, classical cipher-based command encryption, activity monitoring, and intrusion-attempt simulation.

This project will help students understand real-world security concepts used in aerospace communication. Academically, it strengthens applied cryptography and access-control concepts. Industrially, it highlights how critical-infrastructure organizations secure mission-critical systems

2. Background and Justification

Space agencies like NASA, ESA, and ISRO use secure, authenticated channels to operate satellites and spacecraft. Research shows that real-world attacks have targeted mission telemetry, GNSS signals, and satellite command links. Previous academic work focuses on encryption protocols and intrusion detection for aerospace communication. Our project extends these concepts by designing a small-scale simulation of a mission-control security system using introductory information-security techniques such as hashing, classical ciphers, access control, and logging. This justifies its relevance to both learning objectives and real-world scenarios.

3. Project Methodology

We will follow a step-by-step methodology that begins with implementing secure user authentication using SHA-256 hashing, followed by the application of Role-Based Access Control for Admin, Engineer, and Observer roles. The system will encrypt and decrypt mission commands using classical ciphers such as Vigenère, Rail Fence, and Caesar, and will include a secure command interface for sending, viewing, and storing mission commands. Additionally, logging mechanisms and alerts for unauthorized attempts will be implemented. The system will then be tested using simulated cyberattacks, including brute-force attempts and tampered commands, to ensure that each objective is practically demonstrated.

4. Project Scope

The scope of this project includes a secure login system with role-based access levels, classical cipher encryption for mission commands, telemetry and mission-command simulation, logging and intrusion-attempt detection, and a Python-based interface. However, it does not include real satellite communication systems, real-time networking or internet-based communication, hardware integration such as antennas or radios, or advanced cryptographic algorithms like RSA and AES that go beyond the course content.

5. High level Project Plan

Milestone	Description	Duration	Assigned Members
Requirement Analysis	Define all system features	2 days	Whole group
System Design	Flowcharts + architecture	2 days	Saadi & Talha
Authentication Module	Hashing + login system	3 days	Rafay
Encryption Module	Classical ciphers	3 days	Talha
Command & Role System	RBAC + command panel	3 days	Saadi
Logging & Alerts	Intrusion detection + logs	2 days	Whole group
Testing & Report	Test + finalize document	3 days	Whole group

6. References

1. Stallings, William. *Computer Security: Principles and Practice*. Pearson.
2. Pfleeger, Charles. *Security in Computing*. Prentice Hall.
3. NASA Cybersecurity Division – Public Documentation.
4. GeeksforGeeks – Hashing & Cryptography Tutorials.
5. Tutorialspoint – Classical Encryption Techniques.